18 Procurement and Management of Cord Blood Unit for Allogeneic Transplantation

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18.1 Introduction

Umbilical cord blood (UCB) cells for allogeneic use are collected and frozen in more than 160 public CB banks worldwide. More than 800,000 CB units (CBUs) are available for transplantation. In this chapter, we will describe some procedures for cord blood collection, processing, banking, and recommendations on how to choose a single or double UCB unit for transplantation (Querol et al. 2021).

18.2 Collection

During the antenatal period, objective information on the opportunity to donate UCB is given by trained health care providers. After consent, qualified personnel determines donor eligibility to ensure that donation is safe for future recipients. To assess donor eligibility, a donor questionnaire and a medical history review are conducted for identifying risk factors for transmissible and genetic diseases. In addition, infectious disease markers (IDMs) are tested on maternal blood samples obtained within 7 days before or after the collection of the UCB unit. According to local regulations, IDMs are assessed for evidence of infection by HIV-1, HIV-2, hepatitis B, hepatitis C, HTLV-I, HTLV-II, syphilis, and any additional markers required.

Collection technique must not interfere with maternity hospital regular delivery care. Successful collection aims having a high volume that correlates with higher cell contain, avoiding microbial contamination and ensuring traceability. High-quality CBUs are compatible with delayed cord clamping usually up to 1–2 min (Frändberg et al. 2016). After cord clamping, thorough disinfection, venipuncture of umbilical vein to drain by gravity into a primary container containing anticoagulant to avoid clotting. Collection bag and accompanying maternal samples shall be appropriately labeled.

There are two main techniques to collect UCB: before the placenta is delivered (in utero) or after the placenta is delivered (ex utero). Both collection techniques are equivalent regarding quality, providing that collectors are adequately trained and the procedure is validated in each collection center.

After collection, collector will complete a report describing labor characteristics and informing on incidences like the presence of fever or other complications, type of delivery, drugs administered, etc. In case of unexpected adverse reactions during collection, they need to

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be communicated to the competent authority. After collection, it is advisable to follow-up the donor including health questionnaires. Additionally, if any abnormal result is detected during testing, a counseling process should be in place.

Once collection is finished, samples are locally stored in validated conditions before transportation to the processing center. As this facility may be far away from the collection sites, a validated procedure for transportation between these two facilities is needed to demonstrate a controlled environment. Standard operating procedures shall be in place to describe time and temperature of storage and transportation methods. All transportation records shall allow tracking back from the collection site to the UCB bank, and any deviation should be recorded.

18.3 Processing and Banking

18.3.1 UCB Cell Processing

Unrelated UCB unit must arrive at the processing laboratory in time to allow initiation of cryopreservation within 48 h of collection (this time is extended to 72 h for related or directed UCB donations). The decision as to whether a collected UCB unit will be acceptable for processing and banking will be made based on the acceptance criteria specified by the UCB bank. Many banks have further refined their acceptance criteria based on economics and the desire to build an international inventory of high-quality UCB units with very high TNC or shifted toward ethnic minorities. Many UCB banks are now committed to processing and storing only those UCB units with high TNC (e.g., $>150 \times 10^7$ TNC or higher), based on the greater likelihood of these units being used (Saccardi et al. 2016).

Once accepted for further processing, volume reduction of UCB is considered essential to the provision of a high-quality product and cost-effective UCB banking. Reducing the volume of the final product depleting of plasma and RBC allows for storage efficiency in terms of space and cost and, most importantly, reduces the risk of ABO incompatibility and DMSO toxicity to the potential recipient. Despite some cell losses, volume reduction has additional practical and clinical benefits; the process yields RBC and plasma components as waste products that can be used for immediate or future testing, thereby minimizing the loss of the actual stem cell product for testing purposes. Volume reduction is mainly performed by centrifugation using automatic devices and adding some macromolecules to improve buffy coat like hydroxyethyl starch.

The selection of a suitable protocol for cryopreservation of UCB for use in transplantation is critical to optimize the recovery of functionally viable progenitor cells, most of which lie within the $CD34^+$ compartment. Some important considerations that are potential sources of cell damage include the type and concentration of cryoprotectant, the cell concentration, and the cooling and warming rates. UCB units must be stored in freezing bags designed and approved for the cryopreservation of human cells and placed into metal canisters to afford protection during freezing, storage, transportation, and shipping. It is important that after filling, each freezing bag is visually examined for possible leaking and breakage of seals.

UCB units should be cryopreserved using a controlled rate freezer with a validated freezing program. Liquid nitrogen-based tanks have been used to ensure long-term maintenance. Minimizing transient-warming events is very important to maintain viable cells. Stability programs should be designed in order to establish the expiration time of the UCB stored.

18.3.2 Testing and Quality Assessment

Table 18.1 shows release specification for UCB units. Quality assessment is written below:

18.3.2.1 Safety

It is essential that UCB is screened for those infectious diseases which can be transmitted via blood (as described above). In addition, product

Table 18.1 Lists the specification requirements for CBU stored for clinical application, according to appendix V of the seventh edition NetCord-FACT International Standards for Cord Blood Collection, Banking, and Release for Administration (www.factwebsite.org)

Specification requirements for cord blood units stored for clinical administration				
Test	Unrelated specification		Related specification	
	Fresh post-processing sample	Post-thaw attached segment or representative sample prior to release	Fresh post-processing sample	Post-thaw attached segment or representative sample prior to release
Total nucleated cell count	$\geq 5.0 \times 10^8$		Enumerated	
Total nucleated cell recovery	Should be ≥60%		Should be ≥60%	
Total viability	≥85%		≥70%	
Viable CD34 count	$\geq 1.25 \times 10^6$			
Viability of CD34 cells		≥70%		
CFU (or other validated potency assay)[a]		Growth (or positive result for potency)		Growth (or positive result for potency)
Sterility	Negative for aerobes, anaerobes, and fungus		Negative for aerobic and anaerobic bacteria and fungi—OR—Identify and provide results of antibiotic sensitivities	
Donor screening and testing	Acceptable as defined by applicable law and NetCord-FACT standards		Acceptable as defined by applicable law and NetCord-FACT standards	
Identity		Verified		Verified

[a]There should be evidence of potency by CFU or other validated potency assay on a fresh post-processing sample

should be free of microbial contamination (or with an appropriate antibiogram for related uses). Prior to release for administration, each UCB unit must have undergone hemoglobinopathy screening.

18.3.2.2 Identity

At least, HLA-A, HLA-B, HLA-C, and DRB1 loci must be determined using DNA-based methods and result included when listing a UCB unit on the search registries. It is recommended that HLA typing is performed in an accredited laboratory. ABO blood group and Rh type must be reported prior to listing a UCB unit for search. Prior to the release of a UCB unit for administration, it is imperative that HLA identity is confirmed. Ideally, confirmatory typing will be performed on a sample taken from a contiguous segment. HLA typing on maternal blood may also be performed prior to release of a UCB unit. Haplotype matching between maternal donor and infant donor confirms linkage between the two and serves as a secondary confirmation of identity.

18.3.2.3 Purity

UCB unit specifications report total nucleated cells, total nucleated RBC count, and $CD34^+$ cells, and a cell blood count with differential should be performed, with parameters for neutrophils, lymphocytes, monocytes, and platelets defined.

18.3.2.4 Potency

Potency testing to determine the growth potential and viability of progenitor cells in a UCB unit should be performed post-processing (prior to cryopreservation), in addition to being performed on a representative thawed sample prior to release for administration.

18.4 Selecting CBU for Transplantation

The success of the UCB transplantation (UCBT) will depend on the characteristics of the CBU. Tables 18.2 and 18.3 list the recommendation for selection of single and double cord blood units, respectively, for allogeneic transplantation.

Table 18.2 Recommendations for unrelated CBU selection and transplantation[a]

Initial selection of single CBU should be based upon	
(a) HLA matching of the recipient and CBU	
(b) CBU cell dose (TNC ± $CD34^+$) at cryopreservation	
(c) Patient's diagnosis (malignant versus nonmalignant)	
(d) Avoiding CBU containing Ag that matches the specificity of any pretransplant donor-specific anti-HLA Ab in the recipient	
HLA matching	
• *Malignant disorders* (Eapen et al. 2014)	
HLA matching should be based upon allelic typing for HLA-A, HLA-B, HLA-C, and HLA-DRB1 for single CBT	
1. Select an HLA-matched (8/8) CBU. TNC dose should be $>3 \times 10^7$/kg	
2. If an HLA-matched (8/8) CBU is unavailable, select a CBU matched at 7/8 HLA loci. HLA-A or HLA-B mismatches are preferable to HLA-DRB1 mismatches. TNC dose should be $>3 \times 10^7$/kg for 5–7/8 matched units	
3. If a CBU matched at 8/8 or 7/8 HLA loci is unavailable, consider a CBU matched at 5 or 6/8 HLA loci. Avoid mismatches in HLA-DRB1	
4. If CBU 4/8 matched, CBU may rarely be considered as a single CB graft if no other option is available. TNC dose should be $>5 \times 10^7$/kg for 4/8 matched units	
5. CBU 3/8 HLA-matched CBU is not recommended	
• *Nonmalignant disorders* (Eapen et al. 2017)	
1. CBU with HLA 8/8 or 7/8 gives same survival results	
2. CBU with HLA 6/8 and 5/8 gives inferior survival rates and is alternative options	
3. We do not recommend selecting cord blood units with more HLA disparities	
TNC and $CD34^+$cell dose	
• *Malignant disorders*	
Nucleated cell dose[b]	At freezing, minimum TNC dose 3.0×10^7/kg, or
	After thawing, minimum TNC of $2.0–2.5 \times 10^7$/kg
$CD34^+$ cell dose[c]	At freezing, $1.0–1.7 \times 10^5$/kg, or
	After thawing, around $1.0–1.2 \times 10^5$/kg
• *Nonmalignant disorders*[d]	
Nucleated cell dose	At freezing, minimum cell dose 3.5×10^7/kg, or
	After thawing, minimum cell dose 3.0×10^7/kg
$CD34^+$ cell dose	At freezing or after thawing, $>1.7 \times 10^5$/kg
Colony-forming unit assay: This assay is important to evaluate the functional capacity of HPCs after thawing an aliquot or after thawing the product; however, it is difficult to establish a generalized CFU-GM dose due to variations of colony setup and counting between centers	
Other considerations when selecting single CB units	
If many CBUs meeting the criteria above are available, the following factors should also be considered	
• *Use accredited cord blood banks.* For safety, only accredited banks recognized by national and international organizations should be used	
• *ABO compatibility:* ABO compatibility may be associated with improved outcomes, although the data are conflicting	

Table 18.2 (continued)

Initial selection of single CBU should be based upon
• *NIMA*: If the cord banks have the mother's HLA typing, the potential effect of NIMA should be noted in context of clinical trials
• *KIR ligand:* Due to conflicting data, KIR ligand matching should not be used in the selection of CBUs
• *Sex matching:* Sex matching between CBUs and patients in single or double UCBT is not necessary

[a]Based on Eurocord recommendations (Ruggeri 2019, modified)
[b]If the infused TNC dose is 1.0–2.0 × 10^7/kg, the number of $CD34^+$ cells or CFU-GM should be taken into consideration to predict the probability of neutrophil recovery and to discuss the possibility of a second transplant. If both cell doses are lower than recommended, a BM aspirate and chimerism analysis should be performed between days +20 and 28. The absence of engraftment indicates the need for a second transplant; preliminary data shows that haploidentical or double CBT should be considered
[c]Due to variation in counting $CD34^+$ cells, this recommendation should be taken with caution. However, if colonies are not growing, the transplant physicians should consider a second transplant after day +30
[d]For patients with BMF syndromes (aplastic anemia or congenital bone marrow failure states) or hemoglobinopathies, the number of TNC at freezing should be greater than 5 × 10^7/kg

Table 18.3 Additional criteria for double CBU selection

– When a single CBU contains insufficient cells (as specified above), double UCBT is recommended for the treatment of malignant disorders	
– There are currently insufficient data to make recommendations for double UCBT in the treatment of nonmalignant disorders	
HLA matching	
• The recommended indications for HLA matching for single CBU with the recipient for double UCBTs should be used, i.e., the minimum acceptable HLA matching between either CBU and the recipient is 4/6 using low/intermediate typing (antigen) for HLA-A and HLA-B and high-resolution typing (allelic) for HLA-DRB1	
• There is no requirement for intercord HLA matching	
Cell dose	
Nucleated cell dose	At freezing, the sum of both CBUs >3.5 × 10^7/kg
	The minimum cell dose of each unit should be >1.5 × 10^7/kg
$CD34^+$ cell dose	At freezing or after thawing, the sum of both CBUs >1.8 × 10^5/kg
ABO matching	
Two prospective trials (Wagner et al. 2014; Michel et al. 2016) did not demonstrate an advantage of double UCBT in children and young adults with hematological malignancies using a MAC regimen, when a single CBU with an adequate cell dose is available.	

Key Points

- Cord blood donation comprises the following steps: informative and consent process, revision of eligibility criteria, cord blood collection, and finally fresh storage before a standardized transportation to the processing cell lab.
- Cell processing labs require coordination of production and quality control labs to transform the altruistically donated raw material in a medicinal product with predefined specifications that ensure its safety, identity, purity, and potency.
- A public cord blood bank is a stem cell registry that provides ready-to-use banked medicinal products for any patient in need through international networking of stem cell donor organizations.
- Cord blood selection is based on sorting CB units using primary criteria (cell content at cryopreservation and HLA matching, based on allele level matching and including HLA C match) followed by ranking based on secondary criteria depending on the type of disease, disease status, conditioning regimen, patients age, and weight.

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